

5.1.2 Neural Networks (NN)

نیورل نیٹ ورک ایک کمپیوٹر ماڈل ہے جو ڈیٹا سے پیٹرن سیکھتا ہے۔

CS XII

Chapter 5: Application and Impacts of Computing

Lecture 03 of 15

★ Golden

5.1.2 Neural Networks (NN) ★ Golden

LEARN IT

A **neural network** is a computer model that **learns patterns from data**. It is **directly inspired by the human brain**, but it works as a **mathematical computer system** — not as living tissue.

Your brain has billions of nerve cells (**neurons**) that pass signals to each other. A neural network copies this idea with small processing units also called **neurons** or **nodes**. These nodes are **connected** and **pass information from one layer to another**, performing calculations at each step.

A typical neural network has three kinds of layer:

- **Input layer** — receives the raw data (pixels of a face, samples of a sound, marks of a student).
- **Hidden layer(s)** — the “thinking” in the middle. Nodes here combine the inputs and learn features (eye distance, nose shape, an edge in a photo).
- **Output layer** — gives the final answer (unlock / keep locked; “this is a cat”; “pass / fail”).

Input layer



Hidden layer(s)



Output layer

WORKED EXAMPLE

Practical example from the source: face unlock.

Step 1 — Input layer. You look at your phone. The camera captures your face as numbers (brightness of each pixel). Those numbers enter the input layer.

Step 2 — Hidden layers. The system studies face features such as **eye distance** and **nose shape**. Early hidden nodes may detect edges and shadows; later hidden nodes combine those into “this looks like Sara’s face.” It compares them with stored data from when you first set up the phone.

Step 3 — Output layer. If the features match, the phone **unlocks**. If they do not (your brother holds the phone), it stays locked.

The same three-layer story works for an NADRA-style biometric gate at a college: camera (input) → feature matching (hidden) → open the turnstile or not (output).

REMEMBER IT

I-H-O. Input receives, Hidden thinks, Output decides. “In Hidden Out.” A neural network is a **brain-inspired maths machine**, not a copy of a living brain.

COMMON MISTAKE

Mistake: “A neural network is an actual human brain inside the computer.”

Fix: It is only **inspired by** the brain. It is mathematics: numbers, multiplications and additions running on ordinary hardware.

اردو میں سمجھیں

نیورل نیٹ ورک ایک کمپیوٹر ماڈل ہے جو ڈیٹا سے پیٹرن سیکھتا ہے۔ یہ انسانی دماغ سے متاثر ہے مگر ایک ریاضیاتی نظام کے طور پر کام کرتا ہے۔ چھوٹی اکائیاں نیورون یا نوڈ کہلاتی ہیں، جو تھوں میں جڑی ہوتی ہیں: ان پٹ تھ ڈیٹا لیتی ہے، پوشیدہ تھ سوچتی ہے، اور آؤٹ پٹ تھ فیصلہ دیتی ہے — جیسے چہرے سے فون کھولنا۔

EXAM TIP

Very common questions: "What is a neural network?" (2–3 marks), "Name its three layers and explain each with the face-unlock example" (5 marks), and "How is a neural network related to the human brain?"

Full-mark recipe: (1) definition — computer model that learns patterns from data, inspired by the brain, mathematical; (2) neurons/nodes connected across layers; (3) Input / Hidden / Output, each in one sentence; (4) walk the camera → features (eye distance, nose shape) → unlock path. Draw the three boxes if the paper asks for a diagram.

CHECK YOURSELF

1. What are the small processing units of a neural network called?
2. In face unlock, which layer studies eye distance and nose shape?
3. Is a neural network a living brain? Explain in one sentence.

Answers: 1. Neurons or nodes. 2. Hidden layers. 3. No — it is inspired by the brain but works as a mathematical computer system.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A neural network is inspired by:

- a. The brain's networks of neurons (loosely)
- b. OSI cables
- c. Access forms
- d. Bubble sort

MCQ 2. It learns:

- a. Weights that turn inputs into outputs
- b. SQL indexes
- c. TCP windows
- d. Figma auto-layout

MCQ 3. Simple NNs have:

- a. Input, hidden, output layers
- b. Only a pie
- c. Only one Excel cell
- d. No numbers

MCQ 4. This is ★: define, then:

- a. One diagram + two uses
- b. Only careers
- c. Only MVPs
- d. Only keys

MCQ 5. NN vs linear formula:

- a. NN can stack non-linear layers for complex patterns
- b. NN cannot use data
- c. NN is a queue
- d. NN is HCI only

MCQ 6. Digits on paper are a classic:

- a. NN / vision example
- b. Primary key example
- c. Waterfall example
- d. K-map example

Extra short questions

1. Define neural network.
2. Three layer types.
3. Two applications.
4. Need for data.

Extra long question

1. Draw a 3-layer net for pass/fail from 3 marks. Label input, hidden, output. Explain one training idea in words (weights change). ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. A computer model of connected units that learns patterns from data, loosely like neurons.
2. Input, hidden, output.
3. Image recognition; speech; translation.
4. Without enough varied examples the net does not generalise.

Long

1. 3 input nodes, few hidden, 1 output. Training: compare prediction to label, adjust weights. Limit: not a biological brain.

Next lecture: 5.1.3 Deep Learning (DL)